

1. Fig. Q1 shows the block charge diagram of an ideal MOS capacitor. For the following questions, briefly explain how you arrive at your answer.

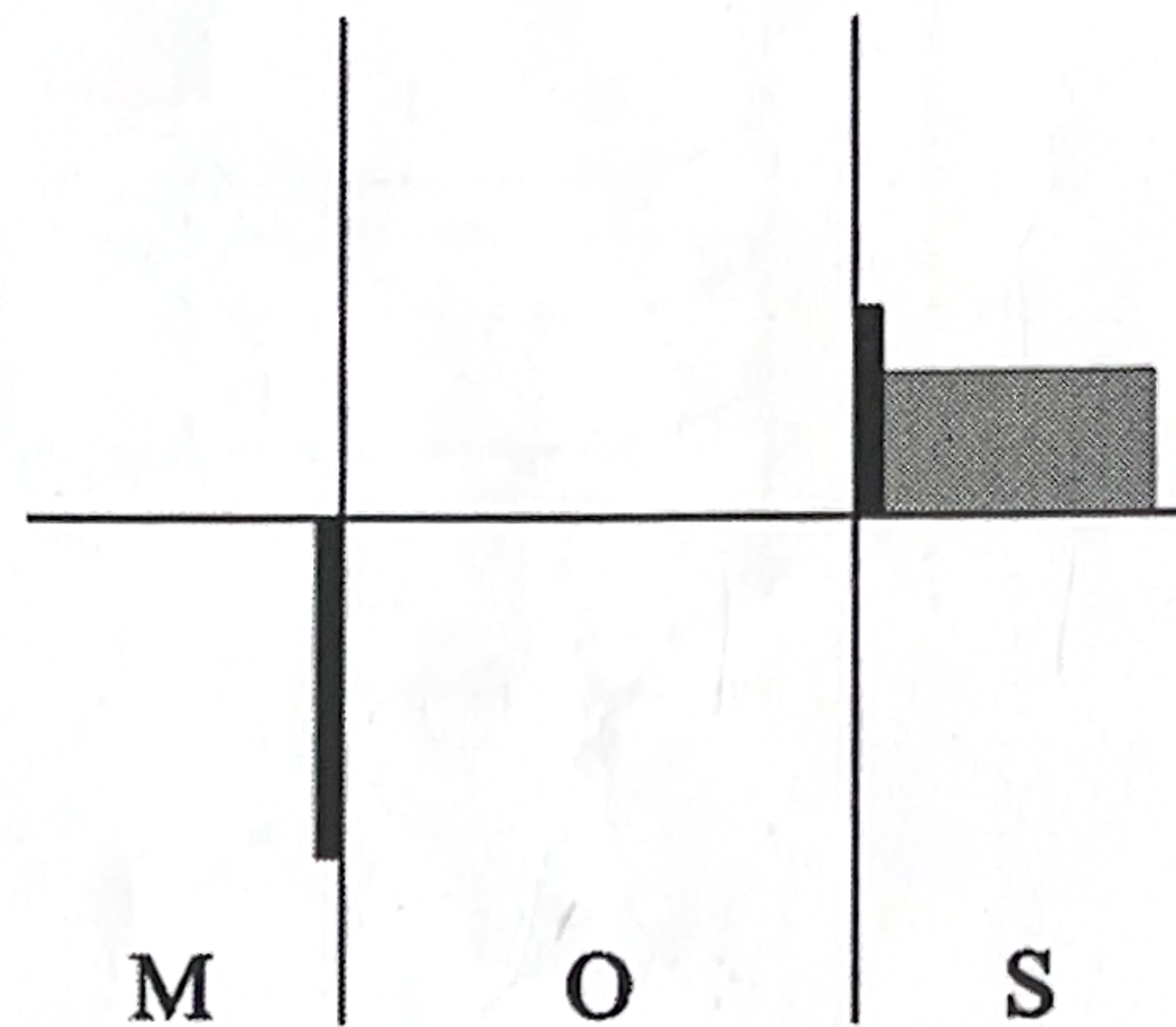


Fig. Q1

- a. Is the semiconductor n- or p-type?
 - b. Is the device biased in the accumulation, depletion, or inversion mode?
 - c. Draw the energy band diagram in the semiconductor region. You may assume that the semiconductor is silicon.
2. An MOS capacitor is fabricated on a p-type silicon substrate, with doping concentration $3 \times 10^{15} \text{ cm}^{-3}$. The oxide thickness is $750 \text{ \AA}$.
- a. Determine the metal-semiconductor work function difference of the MOS structure when the gate is made up of
 - i. aluminium
 - ii. n^+ polysilicon
 - iii. p^+ polysilicon
 - b. Sketch, with reasonable accuracy, the energy band diagrams of the following MOS structures at the stated biasing
 - i. n^+ polysilicon/p-Si at $V_g = 0$
 - ii. p^+ polysilicon/p-Si at $V_g = 0$
 - c. Calculate the threshold voltage of each of the MOS structure in (b).

3. The energy band diagram for an ideal MOS capacitor at $T = 300\text{ K}$ is shown in Fig. Q3. The gate oxide is $0.2\text{ }\mu\text{m}$ thick. The applied gate voltage causes band bending in the semiconductor such that $E_F = E_i$ at the $\text{Si}-\text{SiO}_2$ interface.

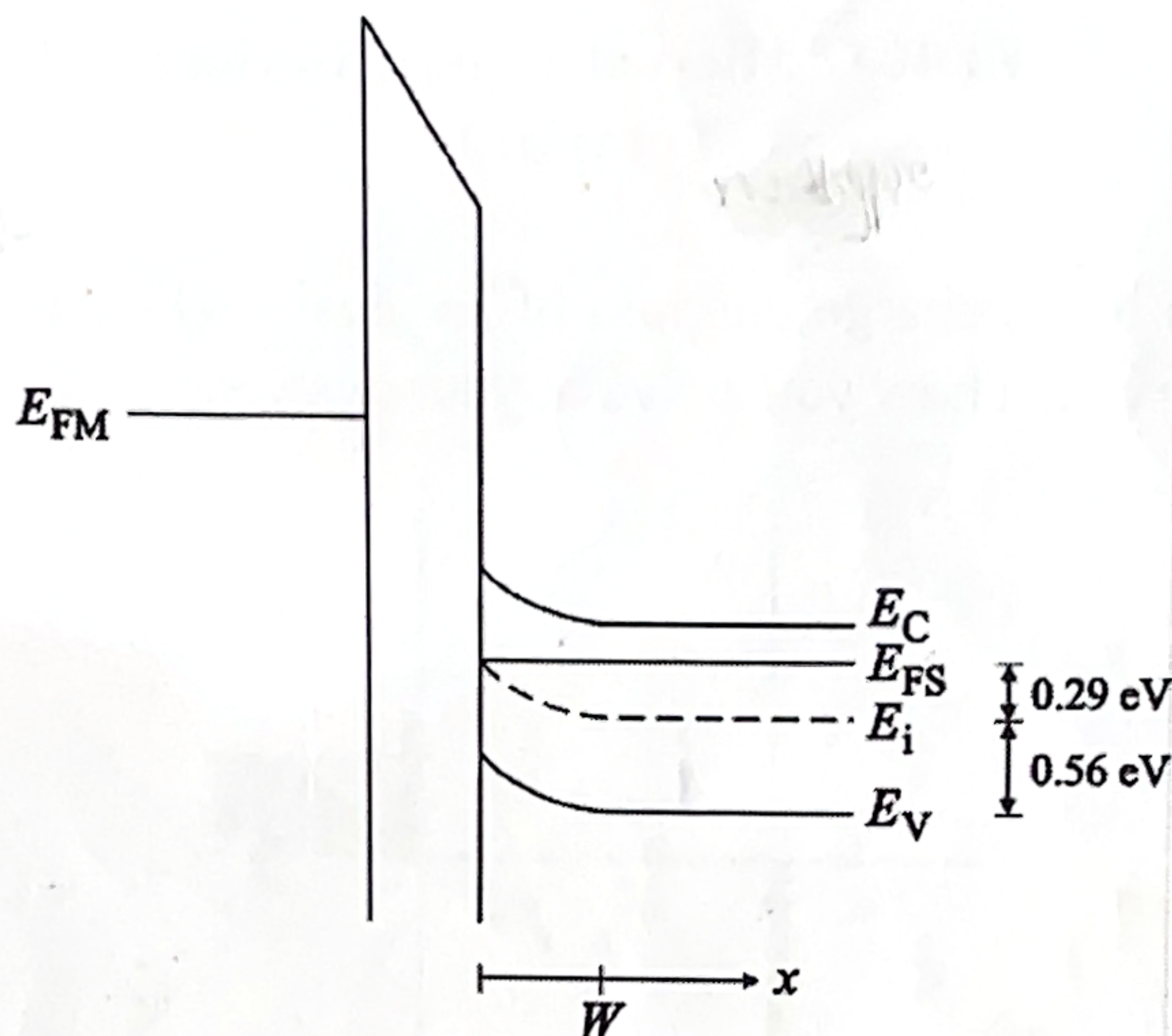


Fig. Q3

- Does equilibrium prevail inside the semiconductor? Explain.
 - Sketch the electrostatic potential and electric field inside the oxide and semiconductor as a function of position.
 - Determine the gate voltage applied.
 - What is the normalized small-signal capacitance, C/C_{ox} of the MOS capacitor at the pictured bias point?
 - Draw and label the energy band diagram of the MOS capacitor at the onset of inversion.
 - Determine the threshold voltage of the MOS capacitor.
4. An ideal MOS capacitor is fabricated on a p-type silicon substrate. The oxide thickness is $500\text{ }\text{\AA}$. The threshold voltage of the device is 0.8 V . Determine the doping concentration of the p-type substrate.